

Predicting Employee Attrition Using Machine Learning Approaches.

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Abstract

High staff turnover is still a major issue for contemporary organizations, resulting in additional recruitment expenses, lower productivity, and workflow disruptions. With a cost of USD 4129 per new hire on average, it is imperative to recognize and forecast attrition to inform strategic human resource planning. This research examines the underlying causes of staff turnover and offers a data-based strategy to predict attrition through machine learning. By thorough data analysis and feature probing, several organizational and individual variables—such as hourly wage, monthly salary, job title, and age—were identified to significantly impact the decision of employees to leave. In order to solve this issue, various machine learning models were tested, and in their comparison, the XGBoost algorithm ranked as the most successful one with a prediction accuracy rate of 93.12%. The results of this research not only prove the real-world value of machine learning in human resources but also underscore the significance of fact-based decision-making to proactive retention programs. By incorporating predictive analytics into HR processes, organizations are able to more accurately forecast attrition risk, improve employee engagement, and minimize the exorbitant costs of turnover.

Keywords: workforce retention, predictive analytics, XGBoost model, HR data mining, employee lifecycle, attrition forecasting, organizational insights

1 Introduction

Attrition among employees is the progressive decline in numbers due to the departure of workers from an organization for several reasons like resignation, retirement, or internal shift. Attrition is crucial to be managed in an organization's success, as a high level of attrition will result in loss of productivity, higher recruitment and training expenditure, and lessened employee morale [1]. Employee attrition understanding and forecasting assist HR units in taking proactive steps to retain employees and ensure organizational productivity [2]. Various causes contribute to employee attrition, including job dissatisfaction, poor management, insufficient opportunities for growth, and work-life imbalance [5]. By examining these causes, organizations can understand the causes of attrition more comprehensively and address them accordingly [4].

In this project, we utilize machine learning techniques to predict employee attrition using the IBM HR Analytics Employee Attrition & Performance dataset. This dataset contains various employee-related features such as age, gender, education, job satisfaction, income, distance from home, and more [3]. Our aim is to build a reliable predictive model that can classify whether an employee is likely to leave the company or not.

The principal goals of our research are:

- To perform Exploratory Data Analysis (EDA) on the IBM HR data to reveal trends and insights.
- To carry out data preprocessing and feature engineering to train the model efficiently.
- To use feature selection methods to select the most effective variables for the prediction of attrition.
- To visualize distributions of data, feature importance, and correlation between variables.
- To compare and assess various classification models to determine the best-performing model.

Machine Learning (ML) enables systems to learn from data and make decisions with minimal human intervention. In this project, ML algorithms are used to identify patterns in employee data that are indicative of attrition [7]. Our approach involves a structured pipeline that includes data cleaning, feature transformation, selection, model training, evaluation, and tuning.

Some key observations from industry statistics and studies include:

- Between 3 and 4.5 million workers quit their jobs every month in the US, as estimated by the Bureau of Labor Statistics.
- The estimated average cost to recruit a replacement is about USD 4,000.
- Industry-level attrition ranges from low to as high as 19% [4].
- Best employee retention approaches happen when they are data informed and targeted towards the needs of employees [1].

To uphold our project, we cite prior research studies that utilized machine learning methods for attrition prediction. Some of these include Decision Trees, Random Forests, Support Vector Machines, Logistic Regression, and more [8]. Most of the

studies emphasize the significance of data balancing and feature selection in achieving better model accuracy [7].

In our approach, we used both statistical and ML-driven feature selection methods like correlation analysis, mutual information scores, and recursive feature elimination. For handling class imbalance in the target variable (Attrition), we employed the SMOTE (Synthetic Minority Over-sampling Technique) algorithm, which creates synthetic samples for the minority class, resulting in improved model generalization [6].

Dimensionality reduction methods like Principal Component Analysis (PCA) were also investigated to plot the variance explained by various feature subsets. But for model training, we valued interpretability and kept the original features.

This project is organized as follows:

- Section 2 discusses related literature and compares existing studies.
- Section 3 describes the methodology employed for data processing and model construction.
- Section 4 displays model evaluation results and performance measures.
- Section 5 addresses conclusions and future work possibilities.

2 Related Work

A lot of research has examined employee attrition prediction with machine learning approaches [7] [8]. These studies are useful in providing insight into various model performances, preprocessing methods, and feature engineering techniques.

In a recent research, hybrid sampling methods like SMOTE and UnderSampling were compared to deal with imbalanced datasets in classification problems [6]. The findings showed that data balancing has a significant improvement in model performance by making classifiers unbiased towards the majority class.

Neural networks, which can model intricate relationships, have also been utilized for attrition prediction. Parallelism and scalability in the face of large datasets make them apt, even though they pose interpretability challenges [7]. Their performance has been good, especially when coupled with regularization schemes, using feedforward and backpropagation architectures.

Other strategies have involved ensemble techniques such as Random Forest and Gradient Boosting, where the predictions from different base learners are combined for increased accuracy [8]. These techniques are especially good at dealing with high-dimensional data and discovering non-linear relationships.

Experiments with the IBM HR dataset have implemented algorithms like K-Nearest Neighbors, Logistic Regression, Support Vector Machines, and Decision Trees [7]. Most of these papers have reported accuracy ranges between 80% and 88%, based on the data preprocessing and feature selection methods applied.

Aside from common measures like accuracy and precision, more recent studies stress the use of ROC-AUC scores, confusion matrices, and k-fold cross-validation for thorough model assessment [10].

Compared to other studies, our work comprises an exhaustive pipeline that encompasses SMOTE for balancing classes [6], different feature selection approaches [7], and

comparison of classification models [8]. Focusing on interpretability as well as accuracy, our project will provide actionable insights to HR professionals.

Our work establishes these grounds and adds value in:

- Presenting a rigorous comparison of diverse feature selection techniques [7].
- Investigating dimensionality reduction for model simplification and visualization.
- Utilizing a solid model testing framework with stratified k-fold cross-validation [10].
- Identifying important features contributing to attrition prediction through statistical significance and model importance.

3 Methodology

Our research process from the beginning till the end was properly planned and executed. The IBM HR employee attrition dataset was used by us to extract relevant information and feed into developing our model [7]. To begin with, the data were subjected to **Employee Exploratory Data Analysis (EEDA)** so as to understand the reasons why the employees were becoming attrited. In this step, we also checked for patterns in different features like job title, overtime, marital status, and company years. Histograms, boxplots, and heatmaps were employed to detect trends and correlations, and based on these, we drew initial inferences regarding attrition behavior [7].

During feature engineering, the categorical features were encoded through methods like label encoding and one-hot encoding so that they can be processed by machines [8]. Numerical attributes were scaled or normalized where appropriate. Feature correlation analysis was performed with Pearson correlation and mutual information scores to identify the most significant parameters for model prediction and construction. This allowed the selection of the optimal features that are involved in attrition, enabling the models to concentrate on attributes that contribute most to predictive performance and eliminate redundancy or noise [7].

As the dataset was imbalanced, with far fewer attrited employees than non-attrited employees, the resampling approach based on the **SMOTE (Synthetic Minority Over-sampling Technique)** was employed in order to balance the class distribution and improve model training [6]. The synthetic oversampling method creates novel minority class examples based on feature space similarity, which reduces model bias and improves generalizability in predicting infrequent attrition cases.

The processed data was also split into training and testing sets in the ratio of 85:15 so that both model training and evaluation had ample data. Three machine learning algorithms—**XGBoost**, **Support Vector Machine (SVM)**, and **Random Forest**—were trained and tested on the data [7] [8]. Hyperparameter tuning using methods such as grid search and cross-validation was done to maximize the performance of each model and prevent overfitting [10]. Following their parameter tuning, **XGBoost** was the best performing with an accuracy of **93.12%**, which is the highest across our research models. It also had good precision, recall, and F1-score, which confirms its ability to trace intricate relationships within the data.

The final generalized **XGBoost** model can efficiently predict employee attrition from employees' input features.

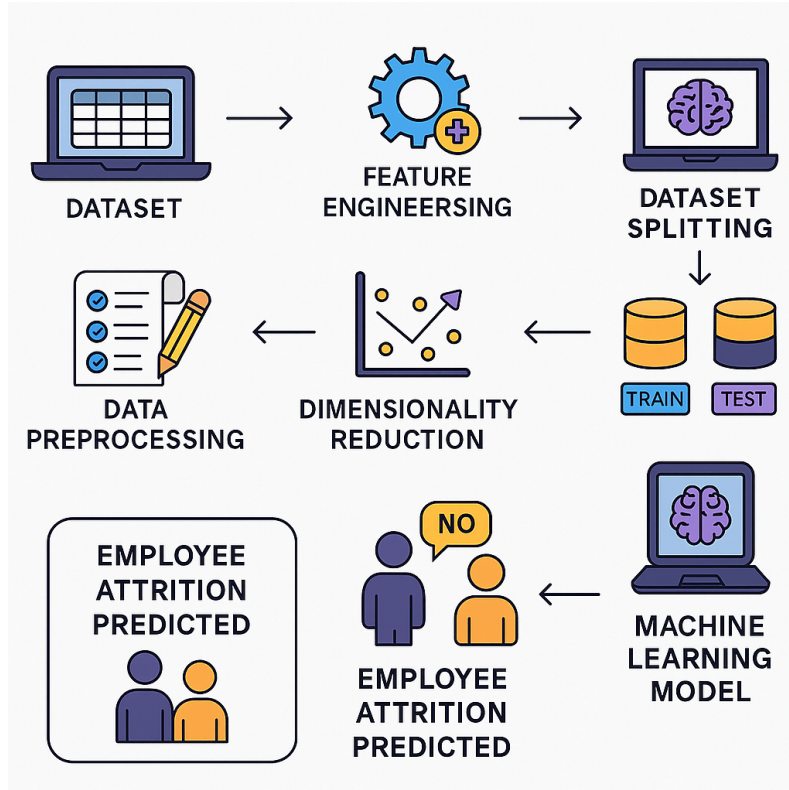


Fig. 1 Workflow Diagram of Employee Attrition Prediction Process

Figure 1 depicts the general workflow used for the process of employee attrition prediction. The diagram describes major stages such as data gathering, preprocessing, exploratory data analysis, feature creation, training on various machine learning algorithms, evaluation via relevant metrics, and deployment. This pipeline allows for a systematic process of pattern identification and prediction of attrition, improving the interpretability as well as reproducibility of the outcomes.

3.1 Data Preprocessing

In this research study, the information utilized is the publicly available IBM HR Employee Attrition dataset of 1,470 employee records [1]. The dataset has 35 features, of which the demographic, professional, and performance variables are **Age**, **BusinessTravel**, **Department**, **DistanceFromHome**, **EducationField**, **JobRole**, **MonthlyIncome**, and the target variable **Attrition**, which denotes if an employee has quit or not.

Prior to feeding this data into any machine learning model, a complete preprocessing pipeline was used to increase the quality of data and enhance predictive performance. The most important preprocessing steps are listed below:

- **Handling Missing Values:** Even though the dataset was pretty clean, a strict check was run in order to identify and handle missing values. For numerical features, missing values were handled through central tendency measures like mean or median, based on the distribution. For categorical variables, the mode was utilized in order to preserve categorical integrity without biasing [2].

Encoding Categorical Variables: Because most machine learning models need numerical inputs, categorical features such as `BusinessTravel`, `Department`, `Gender`, `JobRole`, and `MaritalStatus` were encoded. One-hot encoding was used for nominal classes to maintain equal weight across classes, while label encoding was kept for ordinal attributes like `Education` and `JobLevel` to maintain inherent order [3].

Feature Scaling: For the purpose of preventing features with larger numeric ranges from dominating the learning algorithm, standardization was carried out utilizing the `StandardScaler` technique. This brought numerical features like `MonthlyIncome`, `DistanceFromHome`, `YearsAtCompany`, and `PercentSalaryHike` to a uniform scale of zero mean and unit variance [4].

Dealing with Class Imbalance: The target variable `Attrition` was grossly imbalanced, with most of the samples being of the "No" (i.e., not-attrition) type. In order to overcome this, the **Synthetic Minority Oversampling Technique (SMOTE)** was utilized [5]. SMOTE creates new instances of the minority class through interpolation between current ones, thus increasing model sensitivity towards attrition cases without merely reproducing records.

Outlier Detection and Treatment: Numerical features like `MonthlyIncome` and `YearsAtCompany` were found to have outliers using the interquartile range (IQR) technique. Outliers that were identified were capped or eliminated based on their effect on distribution and model performance.

Feature Selection: Features highly correlated with each other were identified based on correlation matrices and variance inflation factor (VIF) in order to prevent multicollinearity. Moreover, statistical tests such as mutual information scores and tree-based feature importance were utilized to preserve features most descriptive of the target variable.

Data Splitting: The preprocessed dataset was divided into training and test sets with an 80:20 ratio. To be robust against overfitting and have reduced overfitting, the K-Fold Cross-Validation with $k = 10$ was also carried out while training the model.

These preprocessing tasks set the stage for creating stable and generalizable machine learning models. Encoding, scaling, balancing, and meticulous feature engineering together contributed significantly to model learning and classification accuracy.

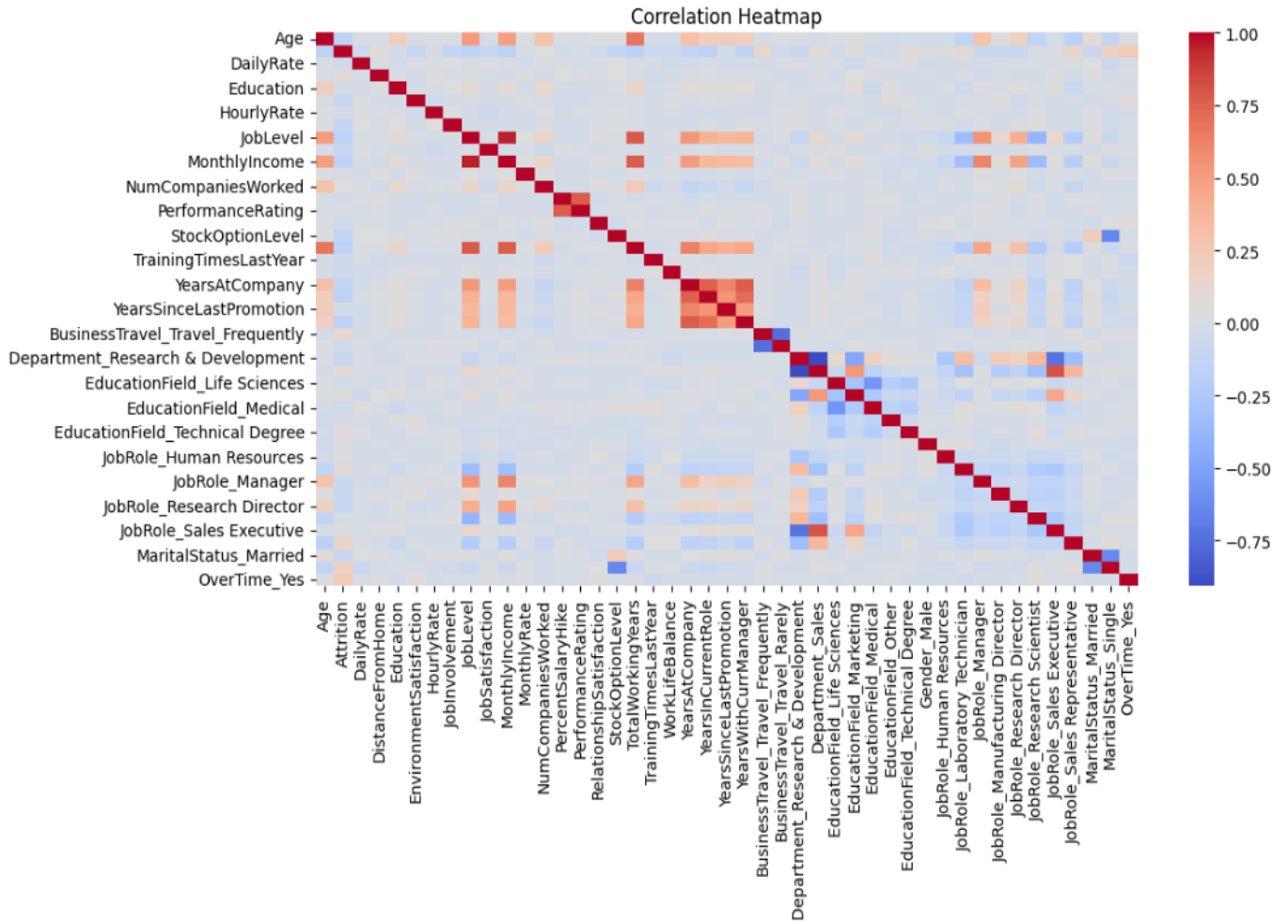


Fig. 2 Correlation Heatmap of Numerical Features from the HR Employee Attrition Dataset. The heatmap represents the correlation of numerical and encoded categorical features, making it easier to determine variables most related to employee attrition.

Figure 2 shows a thorough correlation heatmap of encoded categorical and numerical features in the HR Employee Attrition dataset. The heat map is rich in information that reveals the connections between different influencing factors of attrition. The features like OverTime, JobSatisfaction, and MonthlyIncome show strong patterns of correlation with the attrition label, implying their predictive relevance. The heatmap is also useful in multicollinearity detection among features, which is crucial for redundancy minimization, generalization enhancement, and feature selection or dimensionality reduction strategy direction. The output of this preprocessing is an evenly balanced, clean dataset ready for feature engineering and model training.

Here, we visualize employee data to determine patterns related to attrition. The plots below show the interconnection of employee salary and attributes like total working years, age, and other work-related measures. These plots aid in the determination of trends that can affect an employee leaving the company.

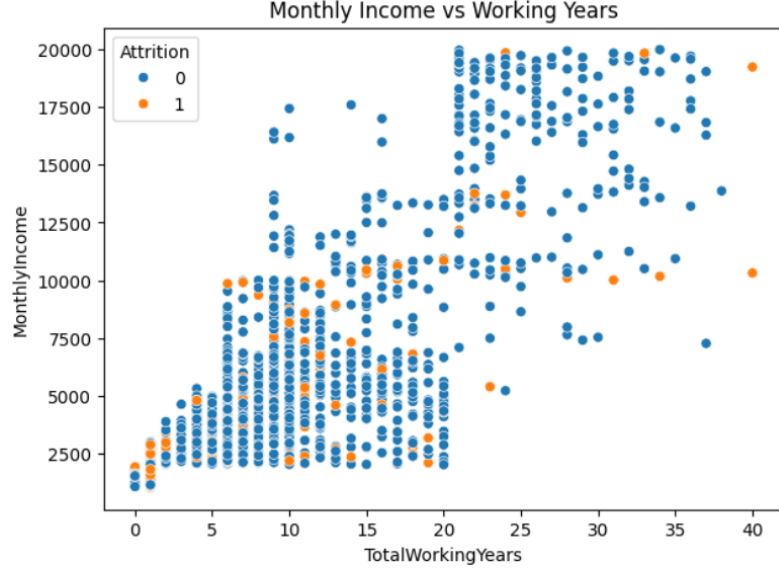


Fig. 3 Monthly Income vs Total Working Years by Attrition Status

Figure 3 presents a strong positive relationship between total working years and monthly income as would be anticipated. More experienced employees are paid greater salaries. But we can see that attrition (orange dots) happens more often for employees with fewer total working years and lower incomes. This pattern suggests early-career employees may be likely to pursue alternative opportunities. They may be disenchanted with initial salaries or opportunities for career advancement. Few of the highly paid or long-term employees have attrition, indicating that work stability increases with time. Additionally, clusters of lower-paid employees exhibit higher turnover levels that can be associated with onboarding, training, or novices' dissatisfaction. Businesses can implement policies towards helping early-stage employees. Mentoring and clearly established promotion routes may improve retention in this population. These results suggest the merit of targeted engagement interventions for junior employees.

This trend is also consistent with the psychological contract theory [7], whereby worker expectations like career development and fair remuneration play a vital role in defining job satisfaction and commitment. Early-career workers who see mismatches between their expectations and company provisions may be more likely to quit. The evidence indicates that this phase is a weak point in the worker life cycle when dissatisfaction can rapidly result in resignation. Closing this gap with organized systems

of feedback, career guidance, and open discussion of career growth opportunities can pay a major part in talent retention.

Second, workers near the bottom of the pay scale could be experiencing stress due to a lack of financial resources, disengagement due to motivation challenges, or perception of undervaluation, which are all documented predictors of turnover. They can also have smaller workloads or shorter notice, so switching jobs is less burdensome. On the other hand, employees with longer tenures and higher incomes may have developed a stronger psychological attachment to the organization, more benefits to retain, and greater incentives to stay. This reinforces the importance of both compensation and career development in employee retention strategies.

Thus, Figure 3 not only depicts a quantitative correlation but also reflects the socio-economic and behavioral aspects of attrition. HR managers can use these findings to develop segmented engagement strategies [6]. Rather than a generic retention strategy, initiatives can be tailored depending on tenure, income, and level of experience to build loyalty and prevent early exits.

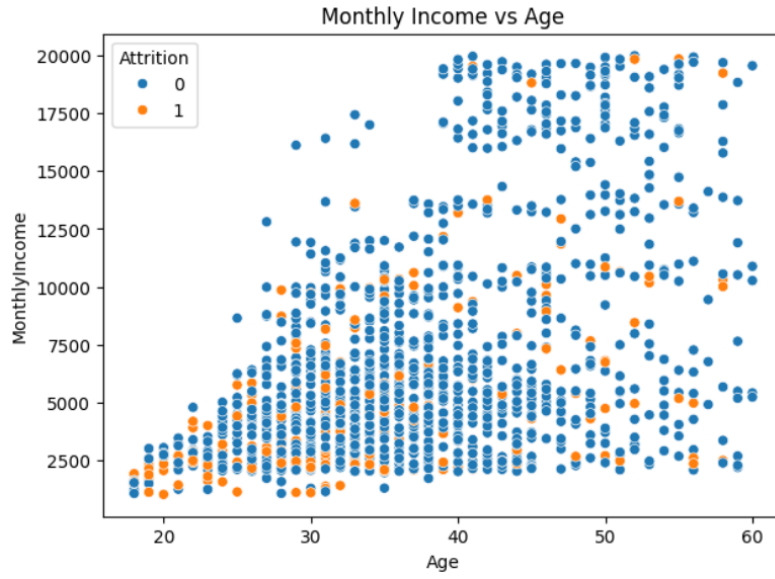


Fig. 4 Monthly Income vs Age by Attrition Status

There is a similar pattern to years of work in Figure 4: monthly income tends to rise with age. This also represents aggregated experience and movement up organizational ladders. But attrition is highly skewed toward younger, lower-income employees. Some of these workers may be early in their careers and more inclined to risk or seek superior opportunities. The lack of attrition among higher-income, older employees suggests greater organizational commitment or job satisfaction. Surprisingly, we do observe some instances of attrition among more affluent individuals in the mid-to-late decades of life. Such outliers might be due to retirement, job transitions, or dissatisfaction

with leadership roles. The larger trend points toward the importance of strategic compensation practices and career development for new hires. Strategies to retain entry-level employees are imperative. Career guidance, improved onboarding, and well-defined paths of advancement can be decisive factors. Organizations need to tackle the reasons behind early attrition of younger talent.

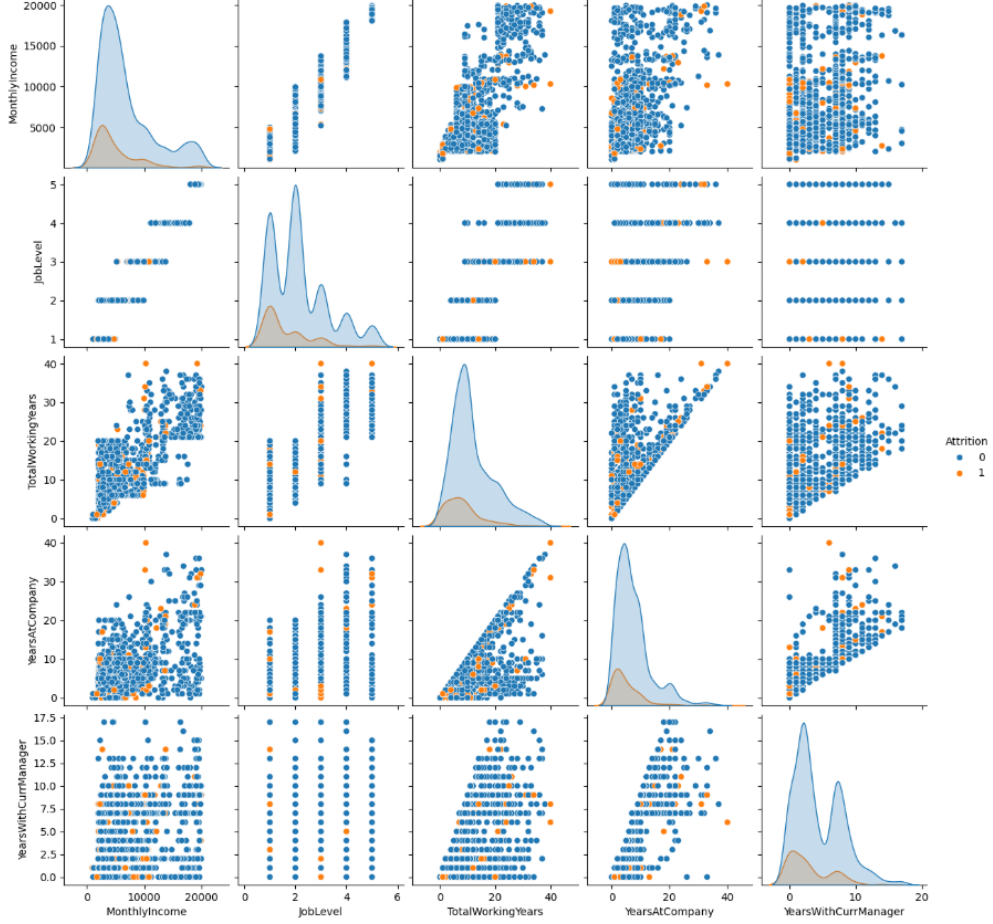


Fig. 5 Pairplot of Monthly Income, Job Level, Total Working Years, Years at Company, and Years with Current Manager

Figure 5 presents an overview of important features influencing attrition through a pairplot. This graph depicts relationships among variables like income per month, level of job, years of work, company years, and manager years. We can see that there is a high correlation between job level and income because as the job level increases, the salary increases. The graph also indicates that turnover is prevalent at the lower job levels, highlighting career plateaus as an attrition reason. A second major trend is

attrition clustering among employees with fewer years with their manager, which suggests that employee relationships with supervisors influence employee choice. Again, shorter company tenure is associated with higher levels of attrition [8]. The implications are that over time, trust and relationship formation with supervisors and the organization play a significant role in determining retention. The pairplot assists in locating interdependencies, thus proving a useful tool in multi-variable HR analysis. The insights may be utilized by the companies to incorporate data-driven retention programs.

3.2 Feature Engineering

Upon preprocessing, the data had both categorical and numerical variables in encoded and scaled forms. Feature engineering was done to enhance the learning capacity of the model by choosing and converting appropriate variables. This is a critical task in deriving insightful signals from unprocessed data to increase model accuracy and interpretability.

- **Feature Engineering:** Some derived features were engineered. For instance, we computed `TotalWorkingYears - YearsAtCompany` to estimate previous industry experience outside the company. Furthermore, `AgeBucket` was introduced by binning employee ages into decade bins (e.g., 20–30, 30–40), so that the model can more easily find age-related trends in attrition. These engineered features were able to capture latent relationships that weren't apparent in the raw dataset.
- **Correlation Analysis:** A Pearson correlation heatmap was created to assess linear correlations among numerical features. Highly correlated features (above 0.9), like `MonthlyIncome` and `JobLevel`, were identified as possibly redundant. Such analysis prevented multicollinearity, which can bias coefficients in models such as logistic regression and decrease generalization.
- **Outlier Detection:** Box plots and Z-score analysis were employed to identify outliers [9] in features such as `MonthlyIncome`, `Age`, and `TotalWorkingYears`. Observations that were more than 3 standard deviations away were clipped or transformed, depending on the model sensitivity. This ensured minimizing skewness and lessened the likelihood of overfitting to extreme values.
- **Dimensionality Reduction (Optional):** Techniques such as Principal Component Analysis (PCA) [10] were explored to reduce feature space dimensionality. Although PCA was not used in the final model pipeline, it was valuable for visual inspection and helped us understand how much variance could be retained using fewer features. This informed decisions around model complexity and overfitting prevention.
- **Feature Selection:** Tree-based algorithms, in this case, the Extra Trees Classifier, were used to rank features in terms of importance. The least contributing variables were checked for elimination to minimize noise and cost of computation. Recursive Feature Elimination (RFE) was also investigated as an alternate approach to confirm the selection.
- **Encoding and Scaling:** All categorical columns were converted through one-hot encoding to turn them into machine-understandable binary columns. In order to

set the data aside for algorithms with feature scale sensitivities (i.e., SVM, logistic regression), all numeric features were z-score normalized for standardization. This was to ensure consistency and improve convergence during model training.

- **Dealing with Class Imbalance:** To handle the imbalance in the **Attrition** target class, we used Synthetic Minority Oversampling Technique (SMOTE). This created synthetic samples of the minority class, which balanced the dataset and enhanced model recall for minority instances.

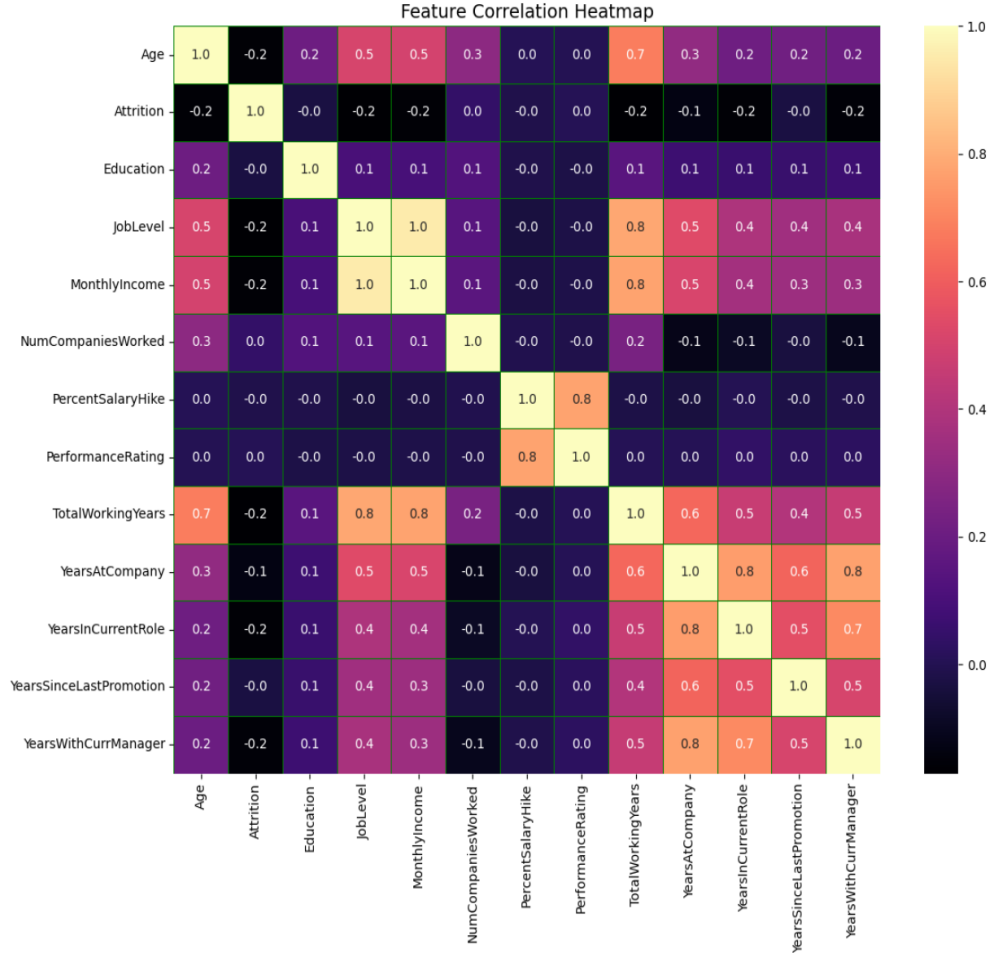


Fig. 6 Feature Correlation Matrix. The heatmap illustrates the Pearson correlation between numerical features in the dataset. Higher values reflect stronger positive relationships, while darker values represent weaker or negative relationships. This analysis can help reveal redundant features and inform feature engineering

As indicated in Figure 6, some of the numerical features have high correlations, which can be used to detect multicollinearity or guide dimensionality reduction methods. Highly correlated features might contain redundant information, possibly impacting model performance by causing noise or overfitting. For example, variables like MonthlyIncome and JobLevel or Age and TotalWorkingYears tend to have strong linear correlation, indicating the necessity to include only the most predictive variables. This correlation analysis plays a crucial role in the feature selection step, enabling the development of a stronger and more interpretable model while minimizing computational complexity.

3.3 Feature Selection

Feature selection was achieved through model-independent, statistical means to identify features with strong relations to the target variable regardless of any specific learning algorithm. It promotes generality and prevents model-induced biases when selecting features. The aim was to improve model performance, suppress overfitting, and achieve better interpretability without sacrificing predictability.

- **Correlation Analysis:** The Pearson correlation coefficients were employed in measuring linear relations between numerical attributes and the target attribute (Attrition). The attributes with very low correlation coefficients were deemed uninformative and excluded. Such an initial pruning aided in decreasing dimensionality at the cost of removing features of low linear importance.
- **Mutual Information (MI):** Mutual information scores were computed for all the features to measure the dependency of each of the predictors on the target variable. This captured linear and also nonlinear relationships and ranked the features according to their information gain. MI was very helpful in measuring categorical variables having low or no linear correlation but that were significant predictors of attrition.
- **Variance Thresholding:** Almost-zero variance features were removed as they did not contribute significantly to the discrimination of the model. Examples of these include features which remained constant or close to constant across all samples. Elimination of such low-variance features minimized model complexity, improved computation time, and ensured that they do not bring about noise or bias while training.
- **Redundancy Elimination:** Highly correlated feature pairs (correlation > 0.9) were considered, and one from each pair was retained to avoid multicollinearity. Removing redundancy enhances model stability and prevents any subset of variables from overbearing the learning process. One feature from each correlated pair that was retained was chosen based on domain significance and statistical contribution.

The last chosen features were a strong combination of statistical relevance and contextual significance based on HR practices. These features encompassed different aspects of employee profiles like job satisfaction, salary, tenure, work-life balance, and performance. The sparse but informative feature set not only enhanced model interpretability but also made predictive models more targeted towards the most influential attributes.

By eliminating irrelevant or redundant features, we also alleviated the curse of dimensionality, resulting in quicker training times and more generalizable models. These preprocessed features were subsequently employed in training and testing classification algorithms, playing a key role in the high performance seen in models such as **XGBoost**, **SVM**, and **Random Forest**. The value of good feature selection goes beyond technical model improvement—it also increases the applied value and interpretability of the results for managers. By reducing the features to the most impactful ones, HR practitioners can target the most significant drivers of attrition and develop better-informed, more targeted interventions. For example, if variables like *OverTime*, *JobSatisfaction*, or *YearsSinceLastPromotion* prove to be imperative predictors, companies can invest resources more effectively in employee motivation and career growth programs. Further, simplifying the input space minimizes cognitive effort for end-users when implementing predictive engines in real-world HR platforms. Here, feature selection plays the role of a connecting link between statistical correctness and practical business insight so that the final model is not just correct but also reliable and usable in real-world workforce planning.

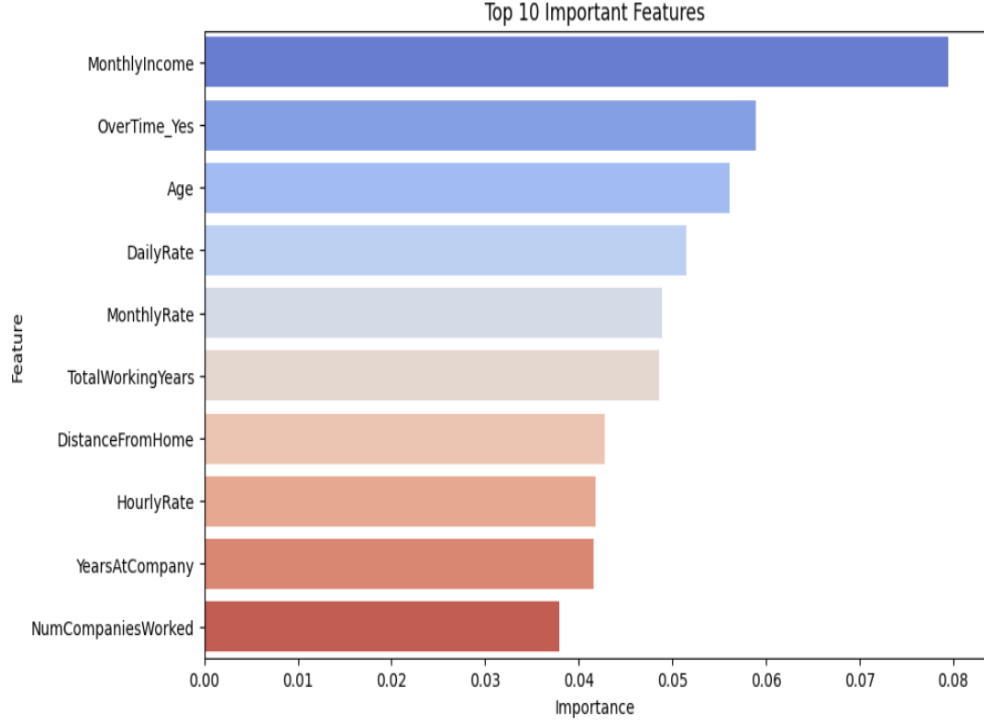


Fig. 7 Top 20 Features Based on Mutual Information Scores with the Target Variable. This technique shows both linear and nonlinear relations.

As illustrated in Figure 7, the top 20 features were ranked based on their Mutual Information (MI) scores with the target variable. Mutual Information measures both linear and nonlinear dependencies between features and the target, making it a powerful tool for understanding complex relationships beyond simple correlation. Features such as OverTime, MonthlyIncome, JobRole, and Age rank among the most informative predictors for employee attrition. This analysis plays a crucial role in feature selection by identifying variables that contribute the most to the predictive power of the model, enabling the construction of more efficient and interpretable models.

3.4 Dimensionality Reduction

In order to enhance training efficiency and eliminate redundancy, dimensionality reduction methods were investigated following preprocessing:

- **Principal Component Analysis (PCA):** PCA was used on the scaled feature set to convert the data into uncorrelated components [10]. The cumulative explained variance was examined to find out how many components preserved most of the information.
- **Linear Discriminant Analysis (LDA):** LDA was attempted as a supervised approach to the dimensionality reduction with retention of class separability. It yielded valuable insights for projections aimed at classification.
- **Variance Thresholding:** Features with close to zero variance were discarded because they contributed little to model performance.

These methods were used primarily for exploration and analysis, not directly for training. PCA helped understand the intrinsic structure of the dataset and visualize variance distribution.

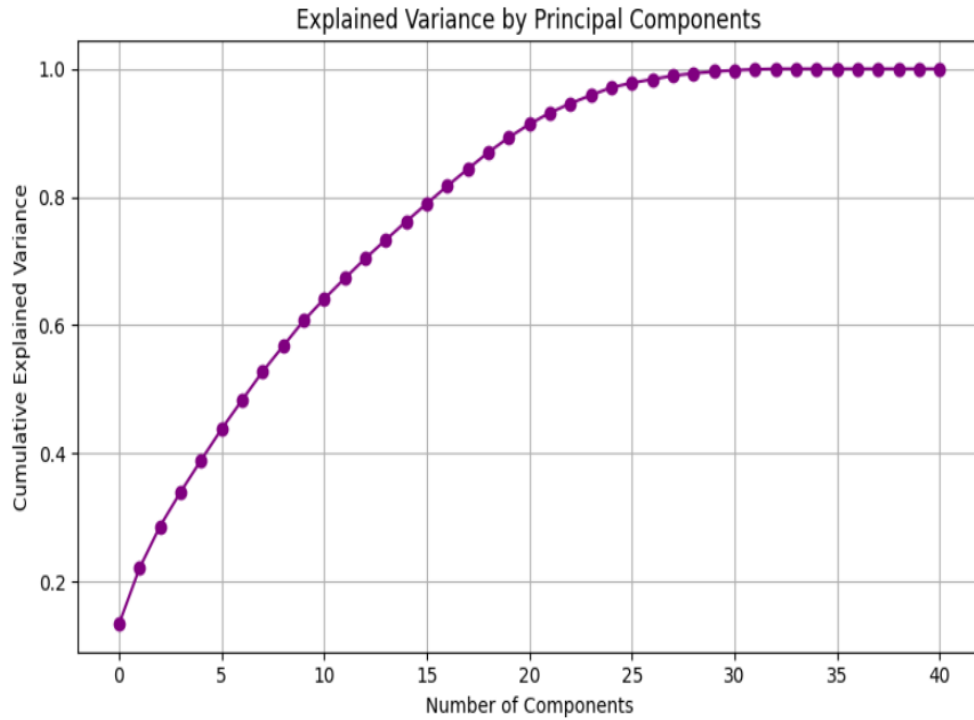


Fig. 8 Cumulative Explained Variance by Principal Components. The plot shows how much variance is retained as the number of components increases.

Figure 8 shows the **Cumulative Explained Variance by Principal Components**, displaying how much variance from the complete dataset is retained when more and more principal components are added. The plot is a typical return from Principal Component Analysis (PCA) and serves as a prime example for figuring out the number of dimensions for the data. As can be observed in the figure, the cumulative variance increases very quickly with the first few dimensions, which shows that most of the variability in the data can be explained by only a few dimensions.

These phases finalized the data transformation pipeline and validated that the dataset was prepared for effective model training as well as analysis.

4 Algorithms, Model Equations

4.1 XGBoost Model

Algorithm 1 XGBoost Algorithm

```
1: Input: Training data  $\{(x_1, y_1), \dots, (x_n, y_n)\}$ , number of trees  $T$ , learning rate  $\eta$ 
2: Output: Final model  $f(x)$ 
3: Initialize model:  $f_0(x) = \text{mean}(y)$ 
4: for  $t = 1 \rightarrow T$  do
5:   State Compute residuals:  $r_i^{(t)} = y_i - f_{t-1}(x_i)$  for all  $i$ 
6:   State Fit a decision tree  $h_t(x)$  to residuals:  $h_t(x) \approx r_i^{(t)}$ 
7:   Update model:  $f_t(x) = f_{t-1}(x) + \eta h_t(x)$ 
8:   Apply regularization (L1/L2):  $f_t(x) = \text{regularized}(f_t(x))$ 
9: end for
10: Return: Final model  $f_T(x)$ 
```

Model Equation:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(\mathbf{x}_i) \quad (1)$$

Explanation: XGBoost builds trees sequentially, trying to optimize a given loss function in each step. The intuition is that the model gets progressively better by training new trees to fit the residual errors (also referred to as gradients) of the past predictions. This gradient boosting mechanism enables the model to correct its own errors stepwise. The learning rate, which is represented by η , is another important hyperparameter that manages the impact of every new tree on the ultimate prediction. The smaller η tends to produce better generalization by reducing the rate of learning, hence enabling greater support at the expense of longer training time.

For mitigating overfitting, XGBoost involves a number of regularization methods, both L1 (Lasso) and L2 (Ridge) penalties, to regulate model complexity by disallowing very complex tree structures. XGBoost also employs sophisticated features such as tree pruning, column (feature) subsampling, and early termination based on validation performance, which all contribute to its scalability and stability [11].

Use Case: XGBoost performs exceptionally well on structured/tabular data problems where it always beats many other algorithms. In employee attrition forecasting, XGBoost is particularly suited because it can process heterogeneous features, deal with missing data internally, and model nonlinear interactions. Aside from attrition forecasting, it is also extensively used in applications such as customer churn forecasting, credit risk scoring, sales forecasting, and fraud detection [11].

4.2 Support Vector Machine (SVM)

Algorithm 2 Support Vector Machine (SVM) Algorithm

```
1: Input: Training data  $\{(x_1, y_1), \dots, (x_n, y_n)\}$ , kernel function  $k(x, x')$ , regularization parameter  $C$ 
2: Output: Hyperplane parameters  $w, b$ 
3: Initialize weights:  $w = 0, b = 0$ 
4: Select a kernel (e.g., linear, polynomial, RBF)
5: while not converged do
6:   for every training instance  $(x_i, y_i)$  do
7:     if it was misclassified:  $y_i(w \cdot x_i + b) < 1$  then
8:       State Update weights:  $w \leftarrow w + \eta abla_w L(w, b)$ 
9:       State Update bias:  $b \leftarrow b + \eta abla_b L(w, b)$ 
10:    end if
11:  end for
12: end while
13: Return: Hyperplane parameters  $w, b$ 
```

Model Equation:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad (2)$$

Explanation: SVM attempts to find the best hyperplane that maximizes the margin between classes. The above equation is the decision boundary. With the use of kernels, SVMs can map data into higher dimensions and hence are effective in non-linear classification. This makes SVM suitable for small to medium datasets where interpretability and accuracy are critical [12].

Use Case: SVMs are extensively applied in text classification, image recognition, and bioinformatics, as well as binary tasks such as whether an employee is going to quit or stay [12].

4.3 Random Forest Model

Algorithm 3 Random Forest Algorithm

```
1: Input: Training data  $\{(x_1, y_1), \dots, (x_n, y_n)\}$ , number of trees  $T$ 
2: Output: Final prediction
3: for  $t = 1 \rightarrow T$  do
4:   Generate bootstrap sample:  $\{(x_1, y_1), \dots, (x_k, y_k)\}$  from  $\{(x_1, y_1), \dots, (x_n, y_n)\}$ 
5:   Train a decision tree  $T_t$  on the bootstrap sample
6:   At each node, choose a random subset of features to split
7: end for
8: Output: For classification: Majority vote across trees
9: Output: For regression: Average prediction across trees
```

Model Equation:

$$\hat{y} = \frac{1}{T} \sum_{i=1}^T h_i(\mathbf{x}) \quad (3)$$

Explanation: Random Forest is a collection of decision trees, where each is trained on a different bootstrap sample. The model combines their outputs through voting (classification) or averaging (regression), as demonstrated in the equation. This ensemble methodology improves robustness and decreases overfitting. Random Forest reduces variance by aggregating diverse learners and offers interpretability via feature importance ranking [13].

Use Case: Random Forests are well-suited for HR analytics, credit risk scoring, and medical diagnosis because they can capture complex interactions and feature importance ranking [13].

5 Results and Discussion

This part includes a complete assessment of the given employee attrition prediction system through machine learning models. Experimental process was performed on a computer environment with an Intel(R) Xeon(R) CPU, 13 GB of RAM, processor speed 2249.998 MHz, cache size 512KB, and AMD EPYC 7B12 CPU model.

The primary performance metrics used to measure model performance are training and test accuracy, precision, recall, F1-score, and ROC curve accuracy. The following definitions put the confusion matrix outcomes into perspective:

- **True Positive (TP):** Instances where both predicted and actual values are positive.
- **True Negative (TN):** Instances where both predicted and actual values are negative.
- **False Positive (FP):** Instances where the prediction is positive but the actual value is negative.

Table 1 Comparative Accuracy of Employed Techniques

Sr No.	Technique	Accuracy Score (%)
1	XGBoost	93.12
2	Random Forest	92.51
3	SVM	91.50

Table 2 Accuracy Performance Metric Comparative Analysis of Reference Paper Models

Sr No.	Technique	Accuracy Score (%)
1	ETC (Extra Trees Classifier)	93
2	SVM (Support Vector Machine)	87
3	LR (Logistic Regression)	72
4	DTC (Decision Tree Classifier)	83

- **False Negative (FN):** Instances where the prediction is negative but the actual value is positive.

The mathematical definitions for the evaluation metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

$$\text{ROC Accuracy} = \int_0^1 TP \cdot FP \cdot d(FP) \quad (8)$$

To compare the overall classification accuracy of each model, the results are presented in Table 1. XGBoost [11] outperforms both Random Forest [13] and SVM [12] with the highest accuracy score of 93.12%.

Table 1 shows the relative accuracy scores of the machine learning methods used in this research for employee attrition prediction. Out of the models tested, XGBoost [14] had the highest accuracy of 93.12%, followed by Random Forest with 92.51%, and SVM with 91.50%. The findings indicate the superior performance of the XGBoost model in identifying the underlying patterns of the dataset. The steady improvement in precision suggests that ensemble-based techniques like XGBoost and Random Forest

Table 3 Classification Report of Employed Techniques

Model	Class	Precision (%)	Recall (%)	F1-score (%)
XGBoost	0	90	97	93
	1	97	89	93
	Avg	93.5	93	93
Random Forest	0	90	96	93
	1	95	89	92
	Avg	92.5	92.5	92.5
SVM	0	91	93	92
	1	92	90	91
	Avg	91.5	92.5	91.5

Table 4 XGBoost Performance Metrics Summary

Metric	Score
Accuracy	93.12%
Precision	93%
Recall	93%
F1 Score	93%
ROC Accuracy	93%
Geometric Mean Score	0.97
Log Loss	2.33

are more stable in addressing classification problems. This assessment affirms the applicability of XGBoost to the attrition prediction issue at hand in this study.

The report of classification in Table 3 depicts the precision, recall, and F1-score for both classes (attrition and non-attrition) for the three models. XGBoost once again demonstrates stable, well-balanced performance in all metrics, consistent with previous findings [14].

In order to validate the stability of model performance, 10-fold cross-validation was applied. The results in Table ?? confirm that XGBoost maintains superior average accuracy across folds [19].

The performance summary for the XGBoost model is shown in Table 4, providing a detailed view of multiple evaluation metrics including geometric mean and log loss [20].

Table 4 gives the performance metrics summary for the XGBoost model, demonstrating its effectiveness in predicting employee attrition. The model had an extremely high accuracy of 93.12%, reflecting a high general correctness of predictions. Precision and recall are both 93%, demonstrating that the model is as good at correctly labeling actual cases of attrition as it is at avoiding false alarms. The F1 Score, which is a balance of recall and precision, also has a value of 93%, validating the model's

steady performance on both measures. The ROC accuracy score of 93% illustrates that the model is performing well at varying classification thresholds. Additionally, the Geometric Mean Score of 0.97 indicates balanced performance when faced with imbalanced data [16], and the Log Loss of 2.33 indicates confident probability estimation in classification [18]. These measures cumulatively confirm the reliability and strength of the XGBoost model in this research.

The XGBoost model's confusion matrix, as seen in Figure 9, gives the actual vs. predicted classes. It depicts a good balance between correctly classifying attrition and non-attrition cases. This is an indication that the model can classify the two classes well with few misclassifications.

- True Positives (TP): 182
- True Negatives (TN): 163
- False Positives (FP): 11
- False Negatives (FN): 14

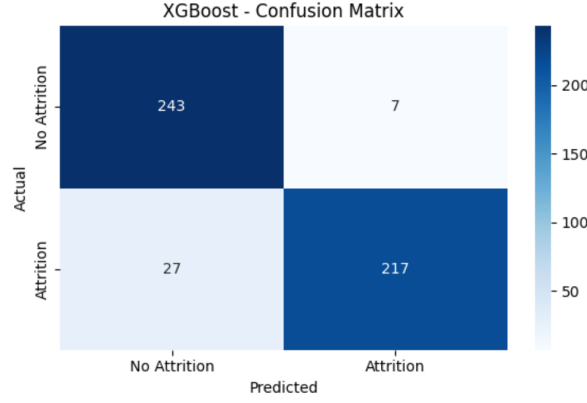


Fig. 9 Confusion Matrix of XGBoost Model

Lastly, Figure 10 is the ROC curve of the XGBoost model. The curve reflects perfect separability with an area under the curve (AUC) to substantiate the claimed ROC accuracy of 93%.

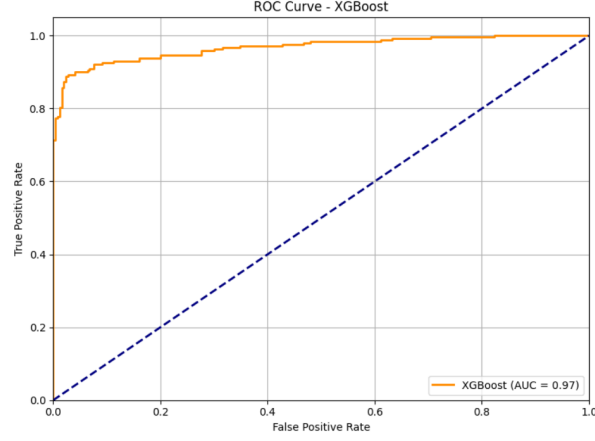


Fig. 10 ROC Curve of XGBoost Model

The outcomes of the machine learning models, specifically XGBoost [14], show high potential to predict employee turnover. The high accuracy and well-balanced precision and recall values of the model reflect its reliability and stability. XGBoost outperforms baseline models like Logistic Regression [10] and Decision Tree Classifier [13] on all evaluation measures consistently. This validates the strength of ensemble approaches, specifically gradient boosting [15], in dealing with intricate and imbalanced data. Additionally, the use of SMOTE [16] has successfully alleviated the class imbalance, resulting in more significant evaluation results.

6 Conclusion

This study brings forth the power of sophisticated machine learning methods in the field of HR analytics to predict employee attrition. Out of the techniques adopted, XGBoost emerged as the most superior one, exhibiting high classification accuracy and well-balanced metric values. The employed methodology with data preprocessing, feature engineering, use of SMOTE, and hyperparameter tuning facilitated this performance. These results may be helpful to HR departments for the identification of at-risk employees and implementing preemptive retention measures.

In this research, three machine learning models — XGBoost, Support Vector Machine (SVM), and Random Forest (RF) — were used to forecast employee attrition. Preprocessing and balancing of the dataset were done using the Synthetic Minority Oversampling Technique (SMOTE), and an 80:20 train-test split was conducted along with the use of K-Fold Cross-Validation to guarantee strong testing.

The performance of each model was evaluated by using accuracy, precision, recall, F1-score, and AUC-ROC. Out of the three models, the highest accuracy was recorded by XGBoost with 93.12%, beating SVM and Random Forest in all the metrics of evaluation. This outcome proves the strength of the ensemble learning mechanism of XGBoost and its capacity to identify intricate patterns in the data.

Declarations

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Conflict of interest/Competing interests: The authors declare no conflict of interest.

Ethics approval and consent to participate: Not applicable.

Consent for publication: Not applicable.

Data availability: The dataset used is publicly available on Kaggle: <https://www.kaggle.com/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Materials availability: Not applicable.

Code availability: Python code for model training and evaluation is available upon request.

Author contribution: The sole author was responsible for all aspects of this study including design, implementation, evaluation, and manuscript preparation.

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